

Numerical Equivalence, Cohomological Equivalence, and Lefschetz Theory for Abelian Varieties over Finite Fields

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To Jean-Pierre Serre, in testimony of our admiration

Abstract

We determine the group of automorphisms generated, inside the endomorphisms of the cohomology of an abelian variety over a finite field, by the Lefschetz operators and by complex multiplications.

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1.

Let k be an algebraic closure of the field $\mathbf{F}_p$ with p elements. We denote by ℓ a prime number different from p . We fix, over k , an isomorphism $\mathbf{Q}_\ell \xrightarrow{\sim} \mathbf{Q}_\ell(1)$.

Let A be an abelian variety over k . We write Z , or $Z(A)$, for the group of cycles with rational coefficients on A , graded by codimension:

$$Z = \bigoplus_i Z^i.$$

On Z^i we have the relations of numerical equivalence and of cohomological equivalence, the latter for étale cohomology with coefficients in $\mathbf{Q}_\ell$. Two cohomologically equivalent cycles are numerically equivalent.

Let A_j be the simple factors of A , let E_j be the center of $\text{End}(A_j) \otimes \mathbf{Q}$, and let F be the compositum, inside an algebraic closure of $\mathbf{Q}$, of the Galois closures of the E_j ; thus F is either totally real or a CM field. If ℓ is a prime number, let $D_\ell \subset \text{Gal}(F/\mathbf{Q})$ be a decomposition group at ℓ ; let $c \in \text{Gal}(F/\mathbf{Q})$ be complex conjugation.

Theorem 1. *If $c \in D_\ell$, then numerical equivalence and cohomological equivalence, for ℓ -adic étale cohomology, coincide for A .*

A less precise theorem, in which the condition on ℓ depended, for a simple variety, on the choice of a maximal field of complex multiplications, was proved in [2].

The following remarks are immediate:

(a) For a given A , the condition of Theorem 1 is satisfied, by the Čebotarev theorem, for a set $\mathcal{L}$ of prime numbers ℓ of nonzero density, either analytic or naive.

(b) If the condition of Theorem 1 is satisfied for A , it is also satisfied for the powers A^n of A . There is therefore a set $\mathcal{L}$, of density > 0 , such that the conclusion is true for every power A^n . This had been observed by Milne [5].

Let $H = \bigoplus_{i=0}^{2g} H^i(A, \mathbf{Q}_\ell) = \bigoplus_{i=0}^{2g} H^i$, where $g = \dim(A)$. Let $D = \text{End}(A) \otimes \mathbf{Q}$.

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The group $D^\times$ acts on H , as does the group, isomorphic to $\mathrm{SL}(2, \mathbf{Q})$, supplied by Lefschetz theory. Both preserve algebraic cohomology classes. Let $G \subset \mathrm{GL}(H)$ be the Zariski closure of the group generated by $D^\times$ and $\mathrm{SL}(2)$. The key point in our proof of Theorem 1 is a description of G , or rather of the group $G_{\mathbf{Q}_\ell}$ over $\mathbf{Q}_\ell$ obtained from it by extension of scalars. In §4 we shall show how this description reduces to the following two cases.

First let E be a supersingular elliptic curve and let $A = E^n$. The Zariski closure of the action of $\mathrm{End}(A)^\times$ on $H \otimes \mathbf{Q}_\ell = \Lambda^\bullet(H^1 \otimes \mathbf{Q}_\ell)$ is the group $\mathrm{GL}(H^1 \otimes \mathbf{Q}_\ell)$. Let $V = H^1 \otimes \mathbf{Q}_\ell$, and let V^* be its dual.

Theorem 2. *The group $G_{\mathbf{Q}_\ell}$ is isomorphic to the group $\mathrm{CSpin}(V \oplus V^*)$ of spin similitudes, where $V \oplus V^*$ is endowed with its natural quadratic form.*

It acts on $H \otimes \mathbf{Q}_\ell = \bigoplus_i H^i(A, \mathbf{Q}_\ell)$ as the sum of its two half-spin representations. In particular, the representation on classes of even degree, respectively odd degree, is irreducible.

At the other extreme, let A be an irreducible variety such that $\mathrm{End}(A) \otimes \mathbf{Q} = D$ is a division algebra, of reduced degree n , over an imaginary quadratic field E ; A has dimension n .

Let $c \in \mathrm{Gal}(E/\mathbf{Q})$ be complex conjugation, and choose an embedding j of E into $\mathbf{Q}_\ell$. The E -module structure on $H^1(A, \mathbf{Q}_\ell)$ gives a decomposition $H^1 = H^{10} \oplus H^{100}$, where $\lambda \in E$ acts on H^{10} , respectively on H^{100} , by multiplication by $j(\lambda)$, respectively by $j(c\lambda)$. Let $V = V_1 \oplus V_2$ be the decomposition dual to $H^1(A, \mathbf{Q}_\ell)$. Over $\mathbf{Q}_\ell$, the Zariski closure of $D^\times$ becomes the group $\mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$.

Endow

$$H^\bullet(A, \mathbf{Q}_\ell) = \bigoplus_{p,q} \Lambda^p V_1 \otimes \Lambda^q V_2$$

with the grading given by $k := p - q$:

$$\mathrm{gr}^k = \bigoplus_{p-q=k} \Lambda^p V_1 \otimes \Lambda^q V_2. \quad (1)$$

The part of degree $(-n)$ is therefore $\Lambda^0 V_1 \otimes \Lambda^n V_2$, of dimension 1, and the part of degree $(-n+1)$ is $V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2$, of dimension $2n$.

Theorem 3. (i) *For an isomorphism of $\bigoplus_k \Lambda^p V_1 \otimes \Lambda^q V_2$ with $\Lambda^\bullet(\mathrm{gr}^{-n+1})$ carrying gr^k to $\Lambda^{\frac{n+k}{2}} \mathrm{gr}^{-n+1}$, the group $G_{\mathbf{Q}_\ell}$ identifies with the group generated by $\mathrm{GL}(\mathrm{gr}^{-n+1})$ acting on $\Lambda^\bullet \mathrm{gr}^{-n+1}$ and by the homotheties $x \mapsto \lambda x$.*

(ii) *The gr^k ($-n \leq k \leq n$) are therefore irreducible, pairwise nonisomorphic representations of G . The subspaces gr^k and gr^ℓ are orthogonal for the cup product if $k + \ell \neq 0$, and are in perfect duality if $k + \ell = 0$.*

2.

We now show how to reduce Theorem 1, and in fact a slightly stronger variant; see §5, to the analysis of the group G .

(a) Let Z_0 be the $\mathbf{Q}$ -vector subspace of $H^\bullet(A, \mathbf{Q}_\ell)$ which is the image of Z , and let Z_1 be the $\mathbf{Q}_\ell$ -vector subspace generated by Z_0 . We shall prove that the intersection form is nondegenerate on Z_1 . Its restriction to Z_0 , which is rational, then has zero kernel: it is nondegenerate and induces a surjection from Z_1 to $\mathrm{Hom}(Z_0, \mathbf{Q}_\ell)$. Hence $\dim_{\mathbf{Q}_\ell}(Z_1) \geq \dim_{\mathbf{Q}}(Z_0)$, the elements of a basis of Z_0 are linearly independent over $\mathbf{Q}_\ell$, and $Z_0 \otimes \mathbf{Q}_\ell \xrightarrow{\sim} Z_1$.

(b) We shall write $\mathbf{Q}_\lambda$ for a sufficiently large finite Galois extension of $\mathbf{Q}_\ell$. It is then enough to verify the nondegeneracy of the intersection form on $Z_\lambda := Z_1 \otimes_{\mathbf{Q}_\ell} \mathbf{Q}_\lambda \subset H_\lambda := H^\bullet(A, \mathbf{Q}_\lambda)$.

(c) Let L be the Lefschetz operator on H deduced from a polarization of A , and let $\Lambda : H^i \rightarrow H^{i-2}$ be the associated operator [4]. The exponentials of the nilpotent endomorphisms L and Λ of H generate a subgroup S , isomorphic to $\mathrm{SL}(2, \mathbf{Q})$. Let G be the Zariski closure, in $\mathrm{GL}(H)$,

of the subgroup generated by S and by $(\text{End}(A) \otimes \mathbf{Q})^\times$. It preserves Z_1 , hence also Z_λ ; the fact that Λ preserves algebraic cycles is due to Lieberman; see [4]. Since G contains the group $\mathbf{G}_m$ corresponding to the grading of $H^\bullet(A)$, coming from the action of $(\mathbf{Q} \cdot 1)^\times \subset \text{End}(A) \otimes \mathbf{Q}$, and also the group corresponding to the same grading shifted by $\dim(A)$, coming from the diagonal matrices in S , it contains the homotheties $x \mapsto \lambda x$.

(d) Suppose that $\mathbf{Q}_\lambda$ is a sufficiently large Galois extension of $\mathbf{Q}_\ell$. We shall check that H_λ is a sum of absolutely irreducible, pairwise nonisomorphic representations W of G , and that the intersection form defines nondegenerate pairings on $W \times W'$, where $W \mapsto W'$ is an involution of the factors. Suppose there is a $\sigma \in \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$ exchanging the factors in duality. Since Z_λ is stable under the Galois group, the intersection form is then nondegenerate; this will finish the proof.

Let E be the center of $\text{End}(A) \otimes \mathbf{Q}$: E is a product of CM fields E_i ; by convention a totally real field is allowed here as a CM field. Suppose that E has the following property, equivalent to the hypothesis “ $c \in D_\ell$ ” of Theorem 1:

there exist a Galois extension $\mathbf{Q}_\lambda$ of $\mathbf{Q}_\ell$ and $\sigma \in \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$ such that (2)

for every i and every embedding $\tau : E_i \rightarrow \overline{\mathbf{Q}_\ell}$, one has $\sigma\tau = \tau \circ c$,

where c is complex conjugation on E_i .

We shall then verify below the existence of a σ exchanging W and W' . Under this hypothesis, Theorem 1 follows.

3.

We recall some linear algebra, referring to [3], which is based on [1]. Let V be a vector space over a field K ; here K is one of $\mathbf{Q}_\ell$, $\mathbf{Q}_\lambda$, or $\mathbf{Q}_\ell$. Let V^* be its dual, and endow $W = V \oplus V^*$ with the nondegenerate quadratic form

$$Q(v + \lambda) = \langle \lambda, v \rangle \quad (v \in V, \lambda \in V^*).$$

Let C be the associated Clifford algebra, the quotient of the tensor algebra $T(V \oplus V^*)$ by the ideal generated by the elements $x \cdot x - Q(x)$, for $x \in V \oplus V^*$. It is a $\mathbf{Z}/2\mathbf{Z}$ -graded algebra [3]; let $C = C^+ \oplus C^-$ denote this grading. Let W act on the exterior algebra $\Lambda^\bullet V$ of V by

$$v \cdot x = v \wedge x \quad (v \in V, x \in \Lambda^\bullet V) \quad \lambda \cdot x = \lambda \rfloor x \quad (\lambda \in V^*, x \in \Lambda^\bullet V) \quad (3)$$

where $\lambda \rfloor x = i(\lambda)x$. The relation defining C is satisfied, and (3) therefore extends to a C -module structure

$$C \longrightarrow \text{End}(\Lambda^\bullet V). \quad (4)$$

The morphism (4) is an isomorphism: see [1], Theorem 2; alternatively, if $V = V' \oplus V''$, then C is the tensor product, in the $\mathbf{Z}/2$ -graded sense, of the analogous algebras C' and C'' , reducing the assertion to the easy case where V has rank one.

The image of C^+ under (4) is the algebra of endomorphisms of $\Lambda^\bullet V$ preserving the $\mathbf{Z}/2$ -grading of $\Lambda^\bullet V$.

The group $C^\times$ acts on the algebra C by conjugation. The subgroup H which normalizes $W \subset C$ respects the $\mathbf{Z}/2$ -grading of C , and is therefore made of even or odd elements. Its action on W defines a map $H \rightarrow \text{O}(W)$ whose kernel is the center of $C^\times$. The subgroup $(C^+)^{\times} \cap H$ of $(C^+)^{\times}$ which normalizes W is the group of spin similitudes $\text{CSpin}(W)$. It maps to $\text{SO}(W)$:

$$1 \longrightarrow \mathbf{G}_m \longrightarrow \text{CSpin}(W) \longrightarrow \text{SO}(W) \longrightarrow 1.$$

Thus $\text{CSpin}(W)$ is the group of automorphisms of $\Lambda^\bullet V$ preserving the $\mathbf{Z}/2$ -grading and the subspace W of $\text{End}(\Lambda^\bullet V)$. Its Lie algebra is the Lie algebra of even endomorphisms T of

$\Lambda^\bullet V$ such that, with W identified as a subspace of $\text{End}(\Lambda^\bullet V)$, one has $[T, W] \subset W$. The representation $\Lambda^\bullet V$ of $\text{CSpin}(W)$ is the sum of the two half-spin representations.

It is clear that $\text{GL}(V)$, acting on $\Lambda^\bullet V$, is a subgroup of $\text{CSpin}(W)$. It lifts, in CSpin , the Levi subgroup of $\text{SO}(W)$ which is the intersection of the two opposite parabolic subgroups preserving the maximal isotropic subspaces V and V^* of W .

The maximal isotropic subspace $V^* \subset W \subset C$ and the line $K \cdot 1$ in $\Lambda^\bullet V$ determine one another: V^* is the subspace of W annihilating 1, and $K \cdot 1$ is the subspace of $\Lambda^\bullet V$ annihilated by V^* . By the same rule, V corresponds to the line $\Lambda^{\max} V$.

The same story repeats for every maximal isotropic subspace of W . The case of interest to us is the one where $V = V_1 \oplus V_2$, and where W is decomposed as the sum of two maximal isotropic subspaces by

$$W = (V_1 \oplus V_2^*) \oplus (V_1^* \oplus V_2).$$

The annihilator of $V_1^* \oplus V_2$ is the line $\Lambda^{\max} V_2 \subset \Lambda^\bullet V$. Fix a generator Ω of it. The inclusion into W of the complementary isotropic subspace $V_1 \oplus V_2^*$ extends to an algebra morphism

$$\Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow C,$$

and

$$I : \Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow \Lambda^\bullet V, \quad \lambda \longmapsto \lambda \cdot \Omega \quad (5)$$

is an isomorphism. The supercommutator with $x \in W$ stabilizes $\Lambda^\bullet(V_1 \oplus V_2^*) \subset C$ and induces a superderivation. For $x \in V_1^* \oplus V_2$, identified with the dual of $V_1 \oplus V_2^*$, this derivation is the interior product $x \rfloor$. Via the isomorphism I , the action of $v \in V_1 \oplus V_2^* \subset W$ therefore becomes exterior multiplication, and the action of $x \in V_1^* \oplus V_2 \subset W$, which kills Ω , becomes interior multiplication.

The exterior product in $\Lambda^\bullet(V_1 \oplus V_2^*)$, transported by I , defines a new product, denoted $*$, on $\Lambda^\bullet V$. If $\Lambda^\bullet(V_1 \oplus V_2^*)$ is endowed with the grading for which V_1 , respectively V_2^* , has degree 1, respectively -1 , then the product is compatible with the grading. The isomorphism I transports this to the evident grading of $\Lambda^\bullet V$, translated by $\dim(V_2)$ so as to place Ω in degree 0. Thus the product $*$ is compatible with the evident translated grading. The natural grading of $\Lambda^\bullet(V_1 \oplus V_2^*)$, for which V_2^* has degree 1, is also compatible with the product. One then obtains on $\Lambda^\bullet V$ the grading gr^k of (1), shifted by $\dim(V_2)$; the element $1 \in \Lambda^\bullet V_2$, of degree 0 for gr^k , is of degree $\dim(V_2)$.

For $v_1 \in V_1$, $v_2 \in V_2$, let us compute $(v_1 \wedge v_2)^\wedge : \Lambda^\bullet V \rightarrow \Lambda^\bullet V$ using the isomorphism (5). We have the diagram

$$\begin{array}{ccc} \Lambda^\bullet(V_1 \oplus V_2^*) & \longrightarrow & \Lambda^\bullet(V_1 \oplus V_2^*) \\ \downarrow I & & \downarrow I \\ \Lambda^\bullet V & \longrightarrow & \Lambda^\bullet V \end{array}$$

and the lower map is a tensor product of maps. If $a \in \Lambda^\bullet V_1$ and $b \in \Lambda^\bullet V_2$, then

$$(v_1 \wedge v_2)(a \wedge b) = (-1)^{\deg a} (v_1 \wedge a) \wedge (v_2 \wedge b).$$

For the isomorphism $\Lambda^\bullet V_2^* \rightarrow \Lambda^\bullet V_2$ defining I , the map $b \mapsto v_2 \wedge b$ is transported to $b^* \mapsto v_2 \rfloor b^*$. It follows that $(v_1 \wedge v_2)^\wedge$ is transported to the endomorphism $v_1 \wedge v_2 \rfloor$ of $\Lambda^\bullet(V_1 \oplus V_2^*)$, which is immediately checked to be a derivation. This derivation preserves $V_1 \oplus V_2^*$.

4.

Computing the group G from §1, after extension to a sufficiently large extension of $\mathbf{Q}_\ell$, is amusing. Here is the story.

(a) Let E be the center of $D = \text{End}(A) \otimes \mathbf{Q}$, a product of CM fields E_i ; D is central simple over E .

(b) After extending scalars from $\mathbf{Q}$ to a sufficiently large $\mathbf{Q}_\lambda$, the algebra E becomes a sum of copies of $\mathbf{Q}_\lambda$, indexed by the set Σ of $\mathbf{Q}$ -algebra morphisms $E \rightarrow \mathbf{Q}_\lambda$. After extending scalars from $\mathbf{Q}_\ell$ to $\mathbf{Q}_\lambda$, $H^1(A)$ becomes $\bigoplus H_\sigma$ for $\sigma \in \Sigma$, where

$$H_\sigma = \{\alpha \in H^1(A, \mathbf{Q}_\lambda) : [z]\alpha = \sigma(z)\alpha \ (z \in E)\},$$

with $[z]$ denoting the action of E on $H^1(A)$ deduced from its action on A . The algebra $D \otimes \mathbf{Q}_\lambda$ is $\prod_\sigma \text{End}(H_\sigma)$.

The group G , acting on $\Lambda^\bullet(\bigoplus_\sigma H_\sigma)$, is generated by the product of the $\text{GL}(H_\sigma)$ and by a “diagonal” $\text{SL}(2)$. Let $L : H^i(A, \mathbf{Q}_\lambda) \rightarrow H^{i+2}(A, \mathbf{Q}_\lambda)$ be the Lefschetz operator, which we identify with a cohomology class in $H^2(A, \mathbf{Q}_\lambda) = \Lambda^2(\bigoplus H_\sigma)$; Lemma 2.1 of [2] implies that

$$L = \sum_v L_v$$

where v runs through the quotient set $\Sigma/\{1, c\}$, complex conjugation acting on Σ through its action on E , and where

$$L_v \in \Lambda^2(H_\sigma \oplus H_{c\sigma})$$

for $v = \{\sigma, c\sigma\}$. The embedding σ may be fixed by c , in which case $L_v \in \Lambda^2 H_\sigma$. Each L_v lies in the closure of the orbit of L under $\prod \text{GL}(H_\sigma)$, and therefore in the Lie algebra of G .

The same arguments give an analogous decomposition of the operator Λ ; the commutation relations are preserved by the components L_v and Λ_v . Consequently, G is also generated by the product of the $\text{GL}(H_\sigma)$ and by one copy of $\text{SL}(2)$, defined over $\mathbf{Q}_\lambda$, for each orbit of c in Σ . Thus G is a product acting on a tensor product.

(c) *The story for $\{\sigma\}$, where $\sigma = c\sigma$.*

This corresponds to a possible $\mathbf{Q}$ -factor of E , hence to a factor of A isogenous to $B = E^n$, where E is a supersingular elliptic curve over k .

Let $V = H_\sigma = H^1(B, \mathbf{Q}_\lambda)$, a vector space of even dimension. Let $W = V \oplus V^*$. The Lefschetz class $L \in \Lambda^2 V$ is an isomorphism from V to V^* . We shall show that the group generated by $\text{GL}(V)$ and by the subgroup $\text{SL}(2)$ associated with L is $\text{CSpin}(W)$ in its spin representation on $H^\bullet(B) = \Lambda^\bullet V$.

One immediately checks that, for its action on $\Lambda^2 V$ by left multiplication, L satisfies

$$[L, W] \subset W.$$

Consequently, L is an element of $\text{Lie}(\text{CSpin}(W))$; it belongs to the Lie algebra of the unipotent radical N of the Levi subgroup defined by $W = V \oplus V^*$ (§3), and is a nonsingular element of it. The conjugates by $\text{GL}(V)$ of the exponential of L therefore generate N . Similarly, Λ and its conjugates generate the Lie algebra of the opposite unipotent subgroup. Hence the group generated by $\text{GL}(V)$ and $\text{SL}(2)$ contains the group Spin ; we have already seen that it contains the homotheties (§2, (c)). Its Zariski closure over $\mathbf{Q}_\ell$ is therefore $\text{CSpin}(V \oplus V^*)$, proving Theorem 2.

(d) *The story for $\{\sigma, c\sigma\}$, where $\sigma \neq c\sigma$.*

In this case $H_v = H_\sigma \oplus H_{c\sigma} = V_1 \oplus V_2$, and $(\text{End}(A) \otimes \mathbf{Q}_\lambda)^\times$ acts on H_v by $\text{GL}(V_1) \times \text{GL}(V_2)$. Write, as in §3,

$$\Lambda^\bullet(V_1 \oplus V_2) \xleftarrow{I^{-1}} \Lambda^\bullet(V_1 \oplus V_2^*).$$

We endow $\Lambda^\bullet(V_1 \oplus V_2)$ with the product $*$, which under I^{-1} becomes the exterior product on the right. We may write ([2, Lemma 2.1])

$$L = \sum e_i \wedge f_i \in \Lambda^2(V_1 \oplus V_2)$$

with $e_i \in V_1$ and $f_i \in V_2$. The isomorphism I^{-1} exchanges the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ with the action on the right-hand side given by

$$\sum e_i \wedge f_i].$$

This is a derivation (§3); indeed, in the bases of V_1 and V_2^* given by (e_i, f_i^*) , it is the sum of the endomorphisms

$$x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$$

of k^2 , hence the endomorphism of k^{2n} thus obtained, extended as a Lie-algebra action to $\Lambda^\bullet(V_1 \oplus V_2^*) = \Lambda^\bullet(k^{2n})$. Therefore the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ leaves the product $*$ invariant in the Lie-algebra sense; that is, it is a derivation.

Moreover the element $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ of $\mathfrak{sl}(2)$ acts on $\Lambda^i V$ by $i - n = \text{gr}^1(\Lambda^i V)$, where gr^1 is the translated natural grading (§3). Since this grading is compatible with the product $*$, the latter is invariant under the Borel subalgebra of $\mathfrak{sl}(2)$. But in a representation of $\mathfrak{sl}(2)$, a vector annihilated by a Borel subalgebra is annihilated by all of $\mathfrak{sl}(2)$. It follows that $*$ is invariant under $\text{SL}(2)$.

Consider the decomposition of $\Lambda^\bullet V$ given by (1):

$$\text{gr}^k \Lambda^\bullet V = \bigoplus_{p+q=k} \Lambda^p V_1 \otimes \Lambda^q V_2.$$

Note that this decomposition is invariant under L and Λ . Indeed, E , via its embedding in $\mathbf{Q}_\lambda$, acts on $\Lambda^\bullet V$. Let $\text{U}(1)$ be the group $\{\lambda \in E : N_{E/F} \lambda = 1\}$, where F is the subalgebra of E formed by the fixed points of c . If $\lambda \in \text{U}(1)$, then λ acts on gr^k by the scalar λ^{p-q} in $\mathbf{Q}_\lambda$; it commutes with L and Λ , and the λ^{p-q} are distinct characters of $\text{U}(1)$. Summarizing:

Lemma 4. *Endow $\Lambda^\bullet V$ with the product $*$ and with the grading (1).*

- (i) *The algebra $\Lambda^\bullet V$ is generated by $\text{gr}^{1-n} = V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2$.*
- (ii) *The product $*$: $\text{gr}^i \times \text{gr}^j \rightarrow \text{gr}^{i+j}$ is invariant under $\text{SL}(2)$, $\text{SL}(V_1)$, and $\text{SL}(V_2)$.*
- (iii) *The grading of $\Lambda^\bullet V$ is preserved by G .*

The image of G in $\text{GL}(\text{gr}^{n-1})$ contains $\text{GL}(V_1)$, $\text{GL}(V_2) \simeq \text{GL}(\Lambda^{n-1} V_2)$, an isomorphism $\Lambda^0 V_1 \otimes \Lambda^n V_2 \xrightarrow{\sim} V_1 \otimes \Lambda^n V_2$ given by L , and its inverse given by Λ . Hence this image is $\text{GL}(\text{gr}^{n-1})$. Property (i) identifies $\Lambda^\bullet V$ with $\Lambda^\bullet(\text{gr}^{n-1})$, compatibly with the action of $\text{SL}(2) \times \text{SL}(V_1) \times \text{SL}(V_2)$. The group generated by $\{zs_1, z^{-1}s_2\}$, where z is a scalar and $s_i \in \text{SL}(V_i)$, inside $\text{GL}(V_1) \times \text{GL}(V_2)$ acts on $\Lambda^\bullet V$ by the action of $\text{GL}(\text{gr}^{1-n})$ described in Theorem 3. The group G is therefore generated by $\text{GL}(\text{gr}^{1-n})$, isomorphic to $\text{GL}(2n)$, and by a one-dimensional torus acting by global homotheties on $\Lambda^\bullet V$; this proves Theorem 3(i). It is clear that, already for the action of $\text{GL}(\text{gr}^{1-n})$, the representations obtained are absolutely irreducible and pairwise nonisomorphic. Finally, if $k + \ell \neq 0$, the spaces gr^k and gr^ℓ are not dual representations, even for the action of $\text{GL}(2n) \subset G$. Since the cup product is nondegenerate, gr^k and gr^ℓ are therefore in perfect duality precisely in the complementary case.

5.

Once the description of G has been obtained, here is the proof of Theorem 1.

Write

$$H_\lambda = H^\bullet(A, \mathbf{Q}_\lambda) = \bigotimes_{v \in \Sigma/c} H_v^\bullet$$

where $H_v^\bullet$ is as in §4. We suppose $\mathbf{Q}_\lambda$ sufficiently large and Galois over $\mathbf{Q}_\ell$.

Under the action of the product, henceforth again denoted G , of the groups described in §4, the space H_λ decomposes as a direct sum indexed by the functions $\sigma \mapsto k(\sigma)$, for $\sigma \in \Sigma$, where

$$k(\sigma) \in \mathbf{Z} \text{ if } \sigma \neq \sigma c; \text{ then } k(\sigma c) = -k(\sigma), \quad (6)$$

$$k(\sigma) \in \mathbf{Z}/2\mathbf{Z} \text{ if } \sigma = \sigma c, \text{ the possible supersingular factor.} \quad (7)$$

In (6), $k(\sigma)$ is given by Theorem 3, extended as in §4 to the case where E is a CM field rather than an imaginary quadratic field. Notice that replacing σ by σc interchanges the spaces V_1 and V_2 of §4, which gives the relation (6).

Let K be the group of functions satisfying (6) and (7). Since the $\mathbf{Q}_\lambda$ -vector space generated by algebraic cycles is stable under G and under the cup product, it corresponds to a submonoid Alg of K . Moreover,

$$\text{Alg} \subset K_0 = \{k : \sum_{\sigma \bmod c} k(\sigma) \text{ is even}\}, \quad (8)$$

$$H_\lambda(k) \text{ is orthogonal to } H_\lambda(\ell) \text{ if } k + \ell \neq 0; \quad (9)$$

$$H_\lambda(k) \text{ and } H_\lambda(-k) \text{ are in duality;} \quad (10)$$

$$\text{Alg is stable under } \text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell). \quad (11)$$

Suppose that the Galois group contains an element acting as (-1) on K . Then, on every $\mathbf{Q}_\ell$ -rational subspace of H_λ which is invariant under G , the duality given by the intersection form is nondegenerate. This applies in particular when, with the notation of §1, complex conjugation $c \in \text{Gal}(E/\mathbf{Q})$ belongs to the decomposition group of ℓ ; hence Theorem 1 follows.

A little less suffices: for every stable submonoid to be a subgroup of K , and therefore for the space of algebraic cycles to contain, together with the space associated with k , its dual associated with $-k$, it is enough that $K \otimes \mathbf{Q}$ have no fixed line under the action of $\text{Gal}(\mathbf{Q}_\lambda/\mathbf{Q}_\ell)$. In other words, if ℓ is unramified in the factors of E , it is enough that Frob_ℓ acting on Σ satisfy:

$$\text{for every } \sigma \in \Sigma, \text{ there exists } a \in \mathbf{Z} \text{ such that } \text{Frob}_\ell^a(\sigma) = \sigma c.$$

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